qualitative analysis presented in the main text, demonstrating that AlphaEdit effectively prevents post-edited LLMs from overfitting hidden representations to the updated knowledge, thereby preserving the original knowledge.

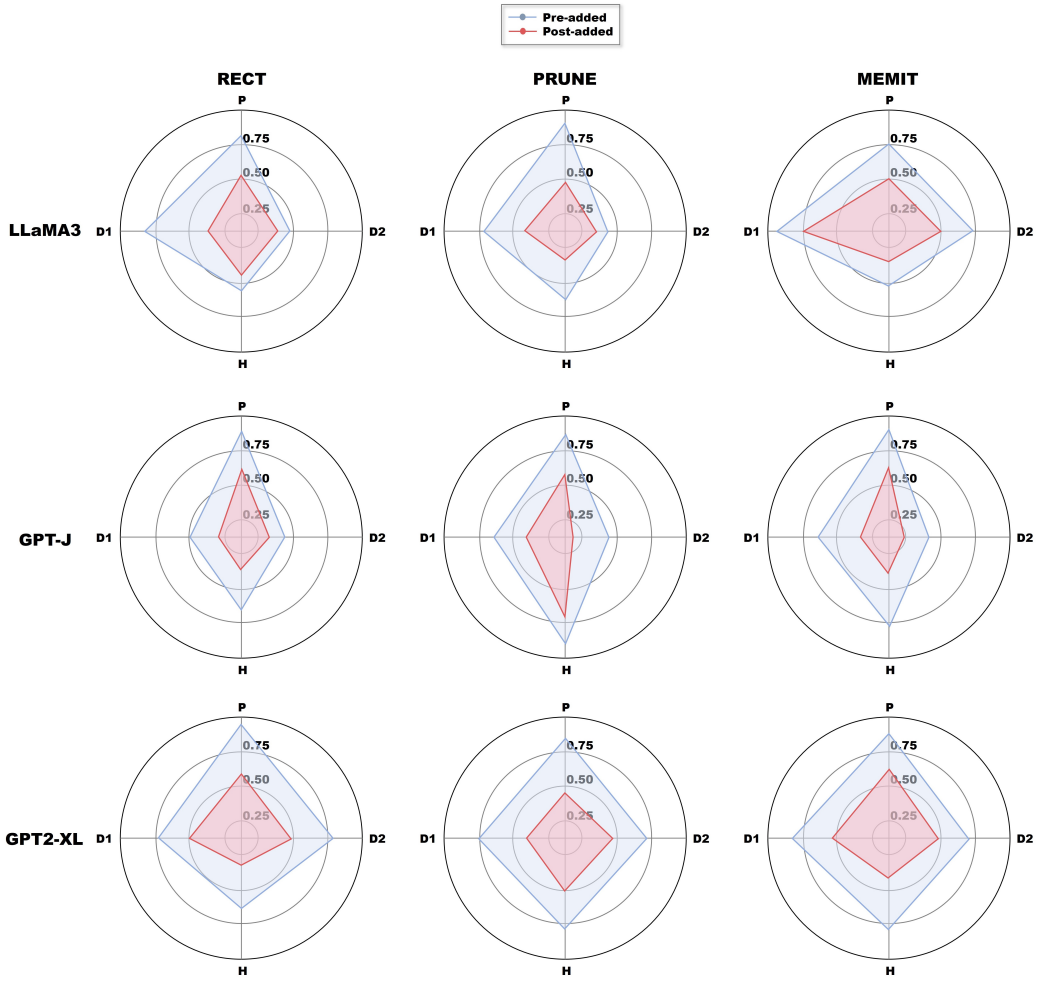

Figure 9: The quantitative results of hidden representation shifts before and after editing. Best viewed in color.

C.4 Editing Facts involving Various Semantics

To gain a deeper understanding of the performance when editing facts involving different semantics, following MEMIT (Meng et al., 2023), we selected several semantics from the Counterfact dataset, each containing at least 300 examples, and evaluated the performance of each baseline methods on these examples (which were evenly distributed across the sequential editing batches). Some of the results are shown in Figure 7 in the main text, with more comprehensive results displayed in Figure 10. In these figures, the horizontal axis represents Accuracy, defined as the average of *Efficacy* and *Generalization* across the 300 examples. For instance, the bar labeled "language," with a height of 98, indicates that out of 1,000 knowledge instances related to language (*e.g.*, "The primary language in the United States is English," or "The official language of France is French"), AlphaEdit successfully edited 98% of them. This metric provides a fine-grained assessment of AlphaEdit's effectiveness across various knowledge domains. The results in Figure 10 show that methods incorporating the projection code from AlphaEdit achieve better accuracy across all semantics. It also indicates that some semantics are more challenging to edit than others. However, even in these more difficult cases, baseline methods enhanced with AlphaEdit's projection code still achieve over 90% accuracy.

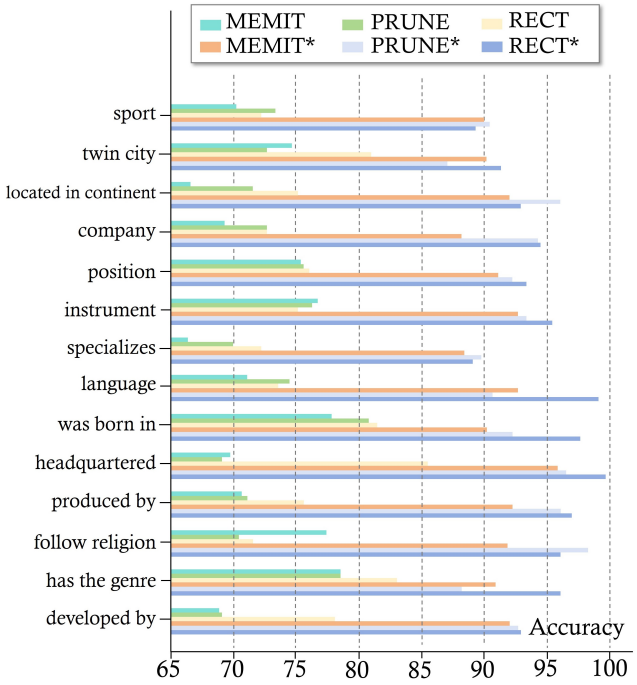

Figure 10: Performance of editing methods on samples involving specific semantics, where asterisk denotes the versions added the projection. Best viewed in color.

C.5 Comparison between AlphaEdit and Memory-based Editing Methods

Although our AlphaEdit mainly focuses on the parameter-modifying editing method, we also hope to explore the advantages of our method compared with some mainstream memory-based editing methods. Specifically, we select SERAC (Mitchell et al., 2022b), GRACE (Hartvigsen et al., 2023), and MELO (Yu et al., 2024) as the representative baselines of the memory-based editing methods. The results are shown in Table 2. According to Table 2 we can summarize that:

- Across all models and tasks, AlphaEdit consistently achieves the highest scores in efficacy (Eff.) and generalization (Gen.). This indicates that AlphaEdit is highly effective at correctly applying the desired edits while maintaining robust generalization to the other knowledge. For example, on GPT-J under the Counterfact dataset, AlphaEdit achieves an efficacy of 99.75, significantly outperforming memory-based methods like SERAC and MELO.
- While AlphaEdit generally achieves competitive scores in specificity (Spe.) and fluency (Flu.), it does not always surpass the memory-based methods. However, we believe this trade-off is reasonable and acceptable because memory-based methods inherently rely on consuming storage space to better preserve existing knowledge.

C.6 Evaluation on Additional Base LLMs: Gemma and phi-1.5

To enhance evaluation diversity, we extended experiments to two additional base LLMs, Gemma (Mesnard et al., 2024) and phi-1.5 (Li et al., 2023). We summarize the results in Table 3. These results demonstrate that AlphaEdit consistently outperforms MEMIT and RECT across key metrics on both the Counterfact and ZsRE datasets. Notably, on Gemma, AlphaEdit achieves the highest fluency (398.96) and consistency (32.91), reflecting its ability to maintain coherence and accuracy. Similarly, on phi-1.5, AlphaEdit excels in efficacy (70.79) and fluency (399.47), showcasing its adaptability to smaller, efficient models. These findings confirm AlphaEdit's robustness across diverse LLM architectures and its capability to deliver high-quality edits while preserving model integrity.

Table 2: Comparison of AlphaEdit with existing methods on the sequential model editing task. *Eff.*, *Gen.*, *Spe.*, *Flu.* and *Consis.* denote Efficacy, Generalization, Specificity, Fluency and Consistency, respectively. The best results are highlighted in bold, while the second-best results are underlined.

Method	Model	Counterfact					ZsRE		
		Eff.↑	Gen.↑	Spe.↑	Flu.↑	Consis.↑	Eff.↑	Gen.↑	Spe.↑
Pre-edited	LLaMA3	7.85±0.26	10.58±0.26	89.48±0.18	635.23±0.11	24.14±0.08	36.99±0.30	36.34±0.30	31.89±0.22
SERAC	LLaMA3	71.21±0.56	61.05±0.39	66.90±0.21	615.72±0.34	20.77±0.13	67.75±0.24	33.96±0.35	22.17±0.15
GRACE	LLaMA3	96.72±0.13	50.14±0.01	**72.23±0.21**	620.43±0.63	23.79±0.23	93.58±0.31	1.03±0.06	31.86±0.12
MELO	LLaMA3	65.29±0.13	58.58±0.32	63.36±0.37	608.98±0.82	22.18±0.04	25.18±0.14	24.14±0.23	30.36±0.75
AlphaEdit	LLaMA3	**98.90±0.10**	**94.22±0.19**	67.88±0.29	**622.49±0.16**	**32.40±0.11**	**94.47±0.13**	**91.13±0.19**	**32.55±0.22**
Pre-edited	GPT-J	16.22±0.31	18.56±0.45	83.11±0.13	621.81±0.67	29.74±0.51	26.32±037	25.79±0.25	27.42±0.53
SERAC	GPT-J	82.28±0.26	58.31±0.34	68.98±0.32	615.92±0.72	28.65±0.17	92.37±0.29	38.21±0.32	25.17±0.25
GRACE	GPT-J	96.50±0.24	50.10±0.01	74.42±0.43	**620.56±0.79**	31.55±0.25	96.54±0.21	0.40±0.02	24.78±0.21
MELO	GPT-J	78.29±0.24	60.52±0.32	66.80±0.52	610.82±0.44	24.31±0.24	82.24±0.07	32.88±0.03	26.65±0.06
AlphaEdit	GPT-J	**99.75±0.08**	**96.38±0.23**	**75.48±0.21**	618.50±0.17	**42.08±0.15**	**99.79±0.14**	**96.00±0.22**	**28.29±0.25**
Pre-edited	GPT2-XL	22.23±0.73	24.34±0.62	78.53±0.33	626.64±0.31	31.88±0.20	22.19±0.24	31.30±0.27	24.15±0.32
SERAC	GPT2-XL	72.25±0.15	58.18±0.23	64.06±0.37	595.35±0.35	27.35±0.12	92.17±0.67	36.57±0.72	20.67±0.22
GRACE	GPT2-XL	98.88±0.28	50.05±0.01	**72.07±0.24**	**620.21±0.49**	28.53±0.15	94.33±0.37	1.59±0.03	**27.63±0.43**
MELO	GPT2-XL	72.62±0.58	53.63±0.42	63.25±0.62	588.57±0.65	23.58±0.33	93.54±0.03	45.25±0.02	23.45±0.24
AlphaEdit	GPT2-XL	**99.50±0.24**	**93.95±0.34**	66.39±0.31	597.88±0.18	**39.38±0.15**	**94.81±0.30**	**86.11±0.29**	25.88±0.21

Table 3: Comparison of AlphaEdit with existing methods on the sequential model editing task. *Eff.*, *Gen.*, *Spe.*, *Flu.* and *Consis.* denote Efficacy, Generalization, Specificity, Fluency and Consistency, respectively. The best results are highlighted in bold, while the second-best results are underlined.

Method	Model	Counterfact					ZsRE		
		Eff.↑	Gen.↑	Spe.↑	Flu.↑	Consis.↑	Eff.↑	Gen.↑	Spe.↑
MEMIT	Gemma	64.68±0.21	60.36±0.30	46.73±0.62	373.94±1.12	22.14±0.31	64.38±0.26	66.12±0.46	**24.52±0.38**
RECT	Gemma	65.17±0.19	57.48±0.64	52.54±0.54	388.77±0.44	23.37±0.39	67.18±0.50	64.12±0.47	20.02±0.47
AlphaEdit	Gemma	**75.21±0.09**	**67.83±0.63**	**52.63±0.49**	**398.96±0.39**	**32.91±0.35**	**75.91±0.42**	**68.12±0.67**	23.50±0.56
MEMIT	phi-1.5	55.71±0.63	56.58±0.78	35.41±0.99	368.57±1.26	19.79±0.31	54.41±0.78	52.47±0.89	**20.98±0.58**
RECT	phi-1.5	58.19±0.73	58.92±0.76	38.46±0.92	362.94±1.44	19.88±0.37	55.15±0.72	53.64±0.83	18.58±0.65
AlphaEdit	phi-1.5	**70.79±0.56**	**65.12±0.88**	**48.96±0.96**	**399.47±0.67**	**25.98±0.48**	**70.02±0.85**	**63.19±0.72**	20.69±0.73

C.7 Evaluation on Expanding Benchmark: KnowEdit, LEME and MQUAKE

In this part, we selected two datasets from the KnowEdit (Zhang et al., 2024d) benchmark, namely wiki_recent and wikibio, for testing. During our experiments, we noticed that some samples within these datasets exhibit abnormally high norms in the hidden states when processed by the LLaMA3-8B-instruct model. These elevated norms are often accompanied by disproportionately large target value norms, which, if forcibly edited, could compromise the model's stability. To mitigate this issue, we implemented an automatic filtering mechanism to exclude samples with excessively high hidden state norms during the continuous editing process. Experimental results are presented in Table 4.

Additionally, we extended our evaluations to two critical datasets, LEME (Rosati et al., 2024) and MQUAKE (Zhong et al., 2023), to test AlphaEdit's performance in more challenging scenarios. LEME focuses on long-form generative tasks, emphasizing consistency, factual correctness, and lexical cohesion. MQUAKE, on the other hand, evaluates the ripple effects of factual updates using multi-hop reasoning questions. Results for these two datasets are summarized in Table 5.

According to Table 4 and 5 we can find that:

- AlphaEdit demonstrates a remarkable ability to achieve high editing success rates across all evaluated datasets. For instance, on the wiki_recent dataset, AlphaEdit achieves an impressive